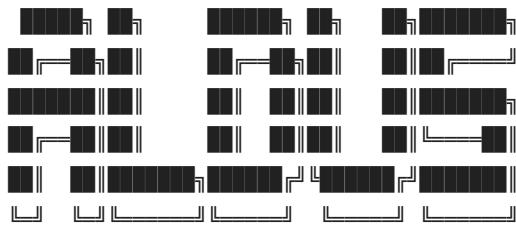


Aldus Test Document



A L E X T I A N



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This document tests every feature Aldus must handle: headings, math, tables, images, code blocks, blockquotes, and lists.

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Headings

Heading 3

Heading 4

Inline Math

The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The area of a circle is $A = \pi r^2$, and the circumference is $C = 2\pi r$.

Einstein's famous equation: $E = mc^2$.

The slope of a line: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Display Math

The Pythagorean theorem:

$$a^2 + b^2 = c^2$$

The sine law:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

A summation:

$$S_n = \sum_{i=1}^n ar^{i-1} = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$$

A limit:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

An integral:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Tables

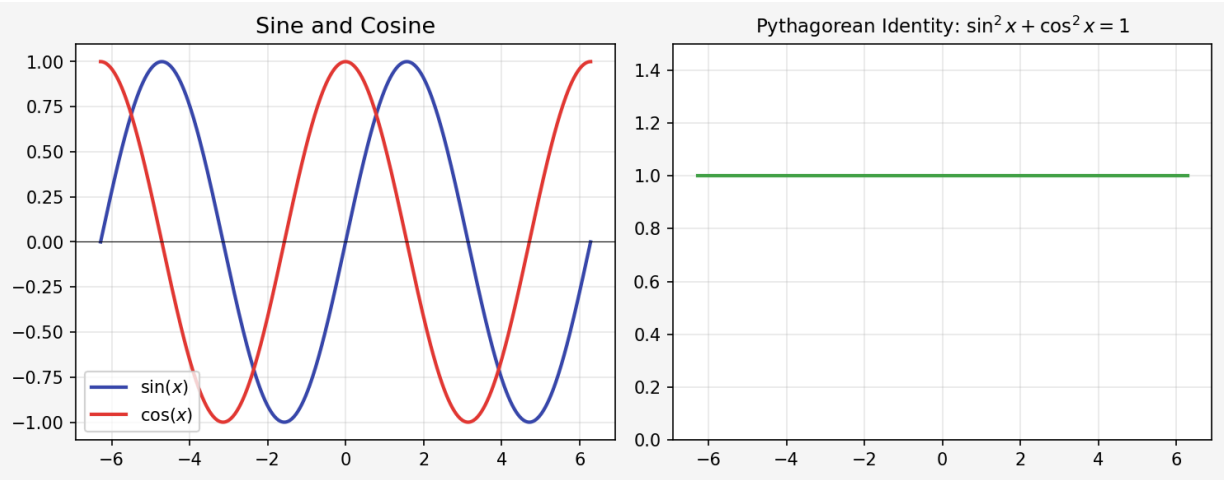
Feature	Status	Notes
LaTeX math	✓	Inline and display
Images	✓	Base64 embedded
Tables	✓	With alternating rows
Code blocks	✓	Monospace font
Blockquotes	✓	Highlighted left border

A math table:

Shape	Area Formula	Perimeter Formula
Circle	$A = \pi r^2$	$C = 2\pi r$
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	$P = a + b + c$
Sphere (surface)	$A = 4\pi r^2$	—

Images

The image below is embedded as base64 — no external file dependency in the PDF.



Code Blocks

Inline code: `f(x) = x^2 + 3x - 4`

A Python code block:

```
def quadratic(a, b, c):  
    discriminant = b**2 - 4*a*c  
    if discriminant < 0:  
        return None  
    x1 = (-b + discriminant**0.5) / (2*a)  
    x2 = (-b - discriminant**0.5) / (2*a)  
    return x1, x2
```

A plain text block:

Step 1: Identify a, b, c from $ax^2 + bx + c = 0$
Step 2: Calculate the discriminant $b^2 - 4ac$
Step 3: If $\text{discriminant} < 0$, no real solutions
Step 4: Apply the quadratic formula

Blockquotes

Key Insight: *The discriminant $b^2 - 4ac$ tells you how many real solutions a quadratic equation has.*

Note: *If $b^2 - 4ac > 0$, there are two real solutions. If $b^2 - 4ac = 0$, there is exactly one. If $b^2 - 4ac < 0$, there are no real solutions.*

Lists

Unordered

- First item
- Second item with **bold text**
- Third item with *inlinemath* : x^2
- Fourth item with inline code

Ordered

1. Identify the type of problem
2. List all known variables
3. Choose the correct formula
4. Solve algebraically

5. Verify your answer

Nested

- Kinematics
 - Displacement: $\Delta x = x_f - x_i$
 - Velocity: $v = \frac{\Delta x}{\Delta t}$
 - Acceleration: $a = \frac{\Delta v}{\Delta t}$
 - Dynamics
 - Newton's Second Law: $F = ma$
 - Weight: $W = mg$
-

Text Formatting

This is **bold text**, this is *italic text*, and this is ***bold italic***.

This is ~~strikethrough~~ text.

Mixed Math and Text

To solve $2x^2 - 4x - 6 = 0$, first divide by 2:

$$x^2 - 2x - 3 = 0$$

Factor the left side:

$$(x - 3)(x + 1) = 0$$

Therefore $x = 3$ or $x = -1$.

We can verify using the quadratic formula with $a = 1$, $b = -2$, $c = -3$:

$$x = \frac{2 \pm \sqrt{4 + 12}}{2} = \frac{2 \pm 4}{2}$$

Which gives $x = 3$ or $x = -1$. ✓

End of test document.

